Hypothesis Testing

↳ It is a statistical method used to make assumptions about a population to claim about something

It is of two types → Null Hypothesis (H_0) [No change]
→ Alternate Hypothesis (H_a) [change]

Ex- New teaching methodology adopted by a teacher improves student's performance.

So,

Acc. to Hypothesis Testing → H_0 = No changes observed in student's performance

H_a = Student's performance is improved

Steps to perform Hypothesis Testing :-

- Assume H_0 & H_a
- Collect sample data
- Compute test statistics ⇝ critical value or p-value
- Choose a significance level (α)
- Compare p-value & α

★ Definitions :-

Null Hypothesis :- No effect, No relationship, No difference b/w groups

Alternate " :- A significant effect or relationship b/w groups

P-value :- A no. calculated from statistical test that helps us to decide whether to reject null hypothesis

↳ Small p-value → reject null hypothesis

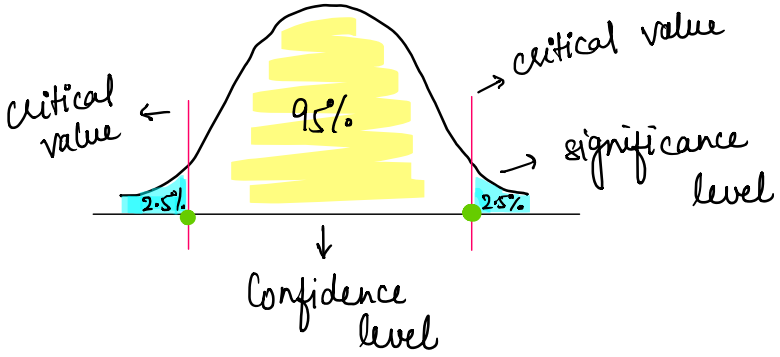
↳ large p-value → reject alternate hypothesis

Significance level (α) :-

- Prob. of rejecting Ho
- It is the opposite of confidence level

If confidence level → 95% or 0.95 then

significance level → 1 - 0.95 = 0.05



Critical Value :- It is the boundary b/w the acceptance or the rejection region.

* Types of Tests :-

① Z- Statistics :- $Z = \frac{X - \mu}{SE}$ where $SE = \frac{\sigma}{\sqrt{n}}$ [used when $n \geq 30$]

② t- statistics :- $t = \frac{X - \mu}{SE}$ where $SE = \frac{s}{\sqrt{n}}$ [used when $n < 30$]

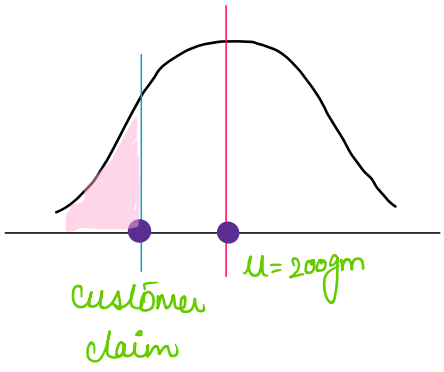
Depending on the above tests, we can observe different kinds of tailed tests :-

Ques. A fast food company says that the burger they create is of 200gm. A customer claims that it is less than 200 gm.

Ques. If [illegible] of 200gm. A customer claims that it is less than 200 gm. What type of test it is?

Sol:- (i) $\mu = 200gm$ ⇝ Ho

$\mu < 200gm$ ⇝ Ha



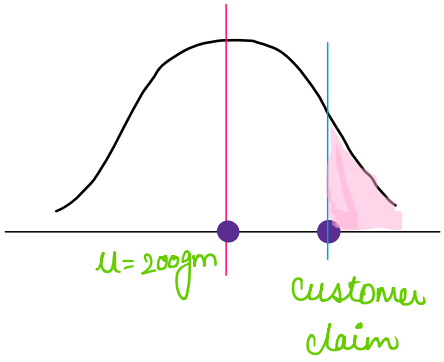
Left-Tailed Test

(ii) If weight is > 200gm then

Ho → $\mu = 200gm$

Ha → $\mu > 200gm$

Right-Tailed Test

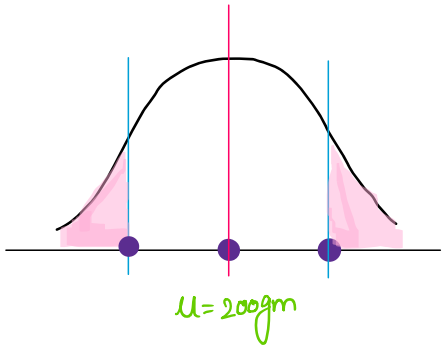


(iii) Weight not equal to 200gm

Ho → $\mu = 200gm$

Ha → $\mu \neq 200gm$

Two-tailed Test



★ P-value Calculation :-

(1) For Two-tailed test :-

$$p\text{-value} = 2 \times [1 - cdf(\text{test statistics})]$$

(2) For one-tailed test :-

$$p\text{-value} = 1 - cdf(\text{test statistics})$$

NOTE:- If p-value < α → reject Null Hypothesis

NOTE:- If p-value $< \alpha \rightarrow$ reject Null Hypothesis

If p-value $> \alpha \rightarrow$ accept Null Hypothesis

* Advanced Statistical Tests :-

Categories for statistical tests :-

(1) Univariate Tests :- Analyzing one variable at a time

Ex- Chi Square Goodness of fit, Kolmogorov Smirnov Test, One-sample t-test.

(2) Bivariate Tests :- Analyzing two variables

↓

evaluating the relationship b/w two variables

Ex- Pearson Correlation, Spearman Correlation, Two-sample t-test, Chi-Square test of independence

(3) Multivariate Tests :- Analyzing the association b/w more than two variables

Ex- One Way ANOVA, Kruskal Wallis Test.

* Parametric vs. Non-Parametric Tests :-

↓

used when data follows a specific distribution

Ex- T-Test, ANOVA &

↓

used when data does not follow any specific distribution

[ideal for non-normal data]

Ex- Mann Whitney U-Test,

Ex- T-Test, ANOVA & Pearson Corr.

Ex- Mann Whitney U-Test, Kruskal Wallis Test.

★ Advanced Statistical Tests :-

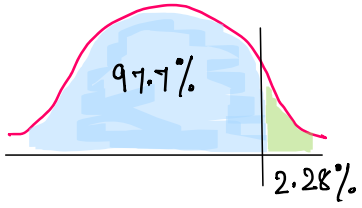
[These tests are conducted in order to find the association b/w multiple variables]

① Z-Test :-

- Used when data is numerical
- Sample size $n \geq 30$
- Population σ is known

Ex- $x = 52$, $\mu = 50$, $\sigma = 6$, $n = 36$, $\alpha = 0.05$

$$Z\text{-score} = \frac{x - \mu}{\sigma/\sqrt{n}} = \frac{52-50}{6/\sqrt{36}} = \frac{2}{1}$$



$$Z\text{-score} = 2.00$$

↓

0.9772 [value from Z-table]

Area from left side = 0.9772

& Area "right" = 1 − 0.9772

= 0.0228

p-value

Now,

$\alpha = 0.05$ & p-value = 0.0228

compare → p-value $< \alpha$ ⇝ Reject H_0

② Chi-Square Test for Goodness of fit :-

- Used when the data is categorical
- Used to check whether the observed frequency matches the expected

frequency or not

Observed Freq. → what the data is in actual

Expected Freq. → What we expect

- Here, H_0 = Observed freq. matches the expected frequency
  H_a = " " do not match the " "

③ Kolmogorov - Smirnov Test :-

- Used when the data is numerical (continuous)
- Used to check whether a sample follows a specific distribution
- H_0 = Sample follows a normal dist.
  H_a = " does not follow a " "

④ One - Sample t - Test :-

- Used when data is numerical
- Used when there is one sample
- Sample size is small ($n < 30$)
- Population σ is not known
- Used to compare sample mean with the population mean
- $H_0 \Rightarrow \mu_{pop} = \mu_s$
  $H_a \Rightarrow \mu_{pop} \neq \mu_s$

⑤ Pearson Correlation Test :-

- Used to test whether two continuous variables have a linear relationship or not.
- $H_0 \rightarrow$ No linear relationship b/w two variables
  $H_a \rightarrow$ Linear relationship b/w the two variables exists

- H_0 → No linear relationship b/w the two variables
  H_a → Linear relationship b/w the two variables exists

(6) Chi-Square Test of Independence :-

- Used to test whether there is a significant association b/w the two categorical variables.
- H_0 → Two variables are independent
  H_a → Two variables are dependent

(7) Two-Sample t-Test :-

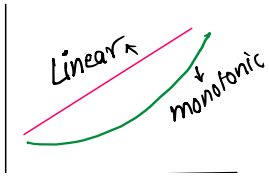
- Used with numerical data
- Used with two independent groups.
- H_0 → Mean of the two groups is same
- H_a → " " " " " is not same

(8) One-Way ANOVA [Analysis of Variance] :-

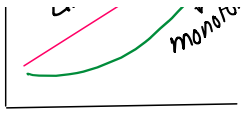
- Used to compare the mean of 3 or more independent groups.
- Used for categorical vs. numerical data
- H_0 : All group means are equal
  H_a : At least one group mean is different

(9) Spearman Corr. Test :-

- Used with numerical data
- Used to test whether there is a monotonic relationship b/w the two variables
- H_0 : No monotonic relationship b/w the variables



- H_0 : No monotonic relationship b/w the variables
  H_a : There is a monotonic relationship b/w the variables



(10) Mann-Whitney U-Test :-

- Non-parametric alternative of t-test
- Used to compare two independent groups
- Data is not normally distributed
- H_0 & H_a → same as t-test

(11) Kruskal Wallis Test :-

- Non-parametric alternative of One-Way ANOVA
- Used to compare 3 or more groups
- Data does not follow a specific distribution
- H_0 & H_a → same as ANOVA